

Design Report: Portable Bluetooth Speaker

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1 System Overview

The system acts as a wireless Bluetooth audio receiver and playback device. The controller is an ESP32 module responsible for receiving and processing incoming Bluetooth audio. The audio pipeline streams data over the I2S digital protocol to dual MAX98357A class-D amplifiers. The final output is driven by dual 5W speakers paired with an LCD display for visual feedback.

2 Hardware and PCB Design

The hardware architecture centers on a custom PCB designed using Altium Designer.

- **Component Integration:** The design connects to the ESP32 module, dual audio amplifiers, LCD display, the ESP32 provides the tactile buttons to control play/pause/skip feature.
- **Signal Integrity:** To ensure high-quality audio, decoupling capacitors and a comprehensive ground pour were implemented as per the manual. The traces connecting the amplifiers were made thicker to handle the current drawn by the speakers and to ensure sound quality.
- **Power System:** The entire system is powered by a portable battery bank connected to the ESP32's built-in USB-C connector. This was done to simplify the design and make the final product more affordable.

3 Firmware Implementation

The software logic driving the hardware was developed as C++ firmware using the Arduino framework.

- **User Interface:** The system features hardware interrupts and debouncing mechanisms to manage the play/pause buttons.
- **Data Handling:** The firmware processes real-time system status and playback metadata, which is subsequently rendered on the integrated display on the ESP32.

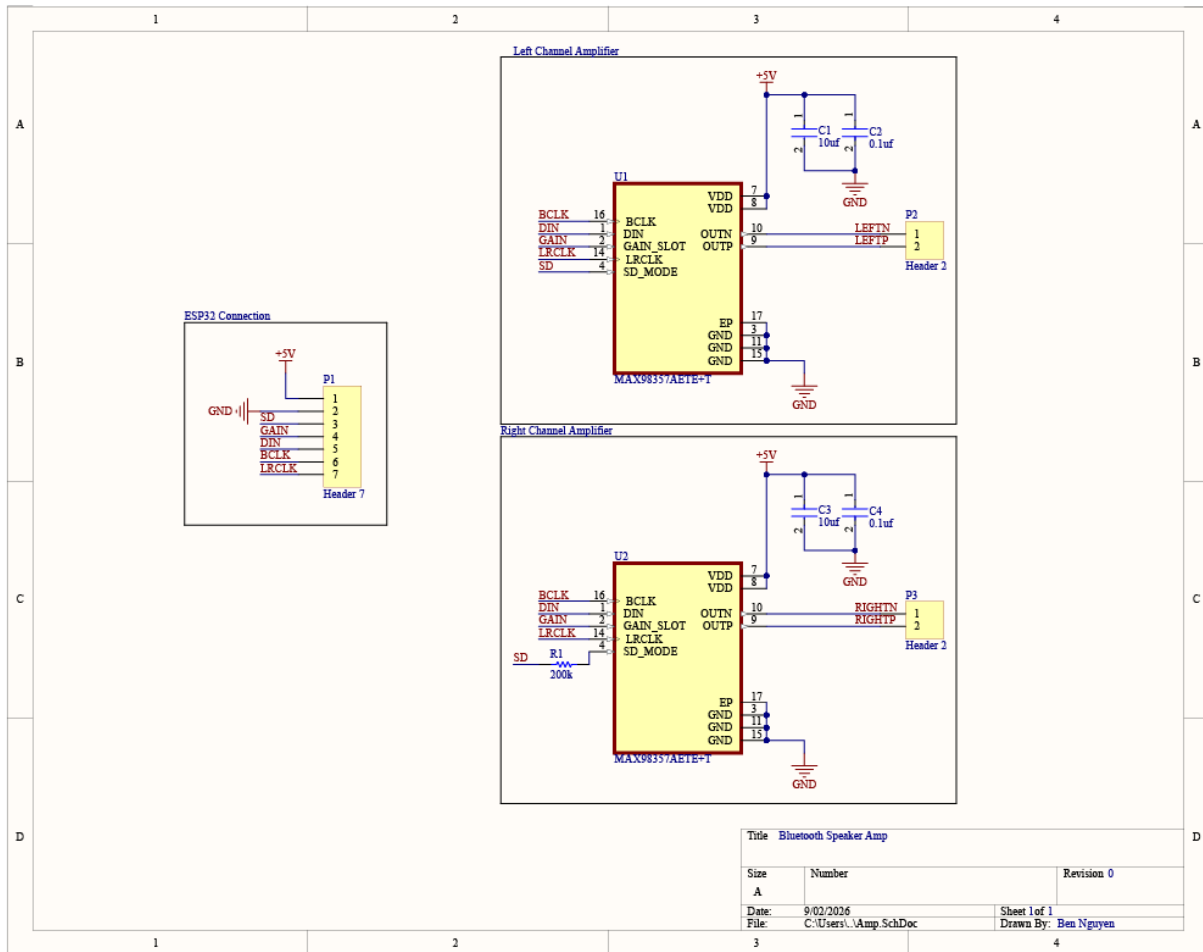


Figure 1: Altium Designer Schematic detailing the ESP32, dual MAX98357A amps, and I2S routing.

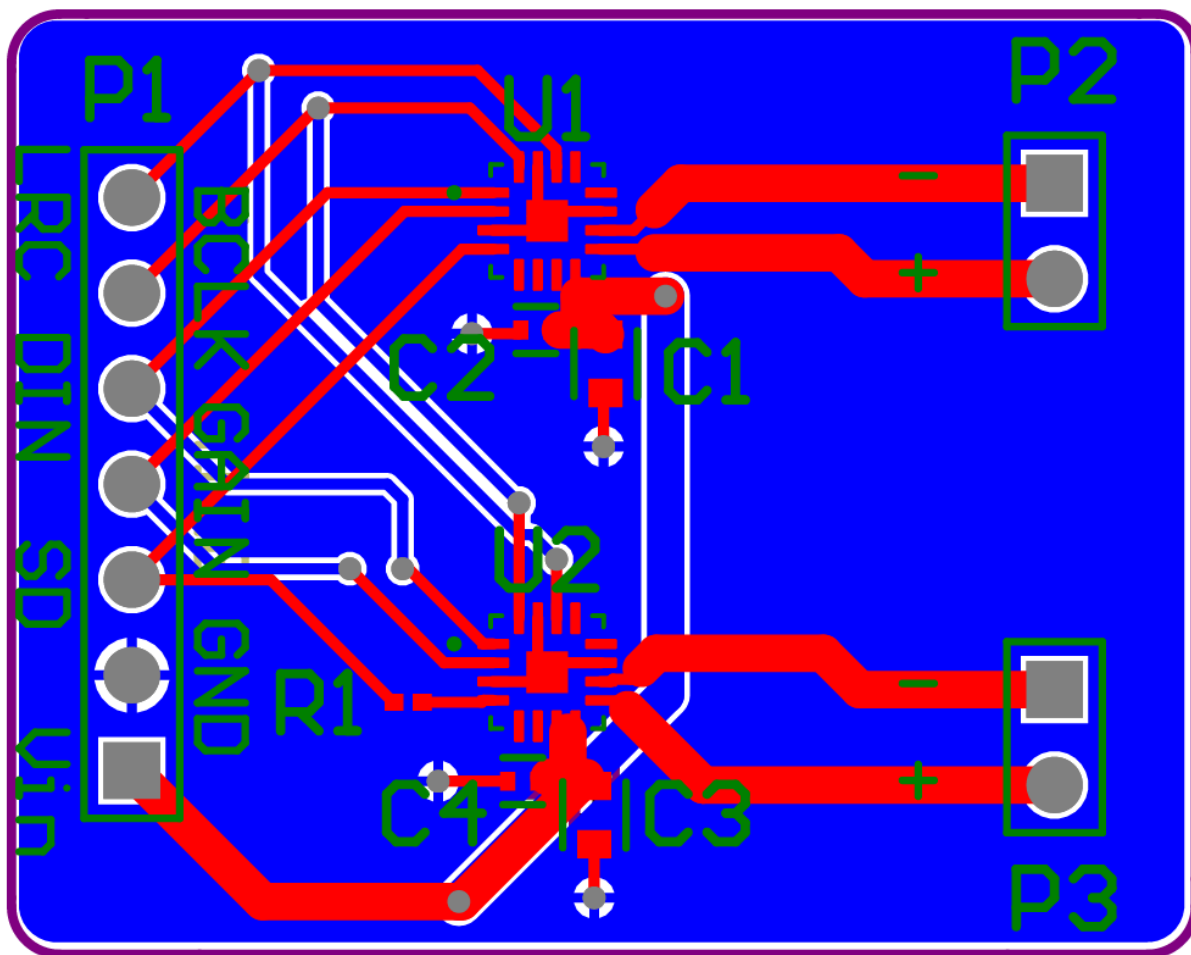


Figure 2: PCB Layout showing the trace runs and the 2 layer design.

4 Mechanical Design and Conclusion

The physical build required a custom housing for electronics and acoustic components.

- **Enclosure:** The enclosure was modeled in SolidWorks and 3D-printed to securely house the PCB, speakers, display, and controls while optimizing acoustic resonance.
- **Testing & Future Improvements:** Initial testing confirmed stable Bluetooth streaming and clean audio output at 5W per channel. Future revisions may include upgraded battery management systems, better acoustic dampening within the enclosure, higher wattage speakers, integrated PCB daughter-board to connect the two systems.